



Warming causes higher bilateral asymmetry in aquatic hemipterans from geothermal streams

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Abstract Increase in temperature associated with climate change may alter community structure and disrupt developmental stability in ectothermic organisms. Using a natural geothermal gradient (20.6–40.3 °C) in Neotropical streams in central-western Brazil, we investigated how warming influences abundance, richness, body size, and fluctuating asymmetry (FA) in aquatic hemipterans. We sampled 18 streams and analyzed 285 individuals belonging to 12 species or morphotypes (Belostomatidae and Naucoridae). Species richness decreased significantly with increasing temperature, whereas total abundance

showed no overall trend; however, species-level responses revealed marked reductions in *Limnoco-ris burmeisteri* De Carlo, 1967 in warmer streams. Adults were less frequent at high-temperature sites, suggesting potential constraints on development or persistence under thermal stress. Contrary to predictions of metabolic theory, no general reduction in body size was detected at the community level, although *L. burmeisteri* exhibited significant decrease in total and standard length along the thermal gradient. In contrast, fluctuating asymmetry increased consistently with temperature, particularly in structures related to locomotion and flight, indicating greater developmental instability under warming conditions. Species-specific analyses further revealed response strategies: *Belostoma anurum* (Herrich-Schäffer, 1848) maintained body size but showed increased

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asymmetry, whereas *L. burmeisteri* exhibited both size reduction and pronounced morphological instability. Our results demonstrate that thermal stress in tropical freshwater systems may reduce species richness and increase sublethal developmental instability, even when changes in body size are not widespread. Incorporating intra-individual morphological metrics, such as fluctuating asymmetry, may improve the detection of early warming effects on aquatic predator communities in Neotropical ecosystems.

Keywords Thermal gradient · Ectotherms · Macroinvertebrate communities · Neotropical streams · Morphometric variation · Predator ecology

Introduction

In recent decades, global temperatures have risen by approximately 1.0 °C, largely due to intensified industrial activity and rapid urbanization (Fischer & Knutti, 2015; Alfieri et al., 2017). Projections suggest further increases of 1.5 °C to 4.5 °C by the end of the century (IPCC, 2023), with pronounced implications in tropical and subtropical regions (Ripple et al., 2023). These temperature shifts profoundly impact ecosystems by altering metabolic rates across biological levels, from individuals to entire communities, contributing to biodiversity loss and jeopardizing ecosystem functioning (Brown et al., 2004; Malhi et al., 2020).

Ectothermic organisms are particularly threatened by temperature changes as their metabolic rates are primarily controlled by environmental temperature (Brown et al., 2004). Given that these groups do not control their body temperature, an increase in temperature intensifies biochemical activities, increasing energy expenditure due to higher enzymatic activity, accelerating chemical reactions, and demanding more energy to sustain biological functions. This temperature-driven increase in metabolic rate can affect ecological processes such as individual and population growth rates, development time, community interactions, and evolutionary rates, with profound impacts on ecosystems (Brown et al., 2004; Savage et al., 2004; Saito et al., 2021).

An approach to study the implications of temperature variations in aquatic ecosystems is to use natural thermal gradients. Geothermal streams, for example,

can have similar size, substrate, location, and seasonality as natural non-thermal streams but with higher temperatures that facilitate the study of the impact of thermal variations on biodiversity and ecosystems (Zivic et al., 2014; Macaulay et al., 2021). In some regions, these systems offer a range of temperatures within the same biogeographic area, providing the opportunity to simulate the effects of rising temperatures without the geographical biases found in comparisons between ecosystems at different locations, e.g., across latitudes (O’Gorman et al., 2019). Yet, the majority of studies are conducted in temperate or boreal geothermal locations, where temperatures typically range from 0 to 25 °C. In contrast, tropical geothermal systems remain poorly studied, despite exhibiting distinct thermal regimes and environmental conditions.

Among the potential effects of temperature changes, we focus on the genetic determination of bilaterality, which may deviate due to environmental changes stressing organisms (Sanseverino & Nesimian, 2008). These deviations suggest constraints in energy allocation for cellular restoration, affecting the organism’s ability to develop bilaterally symmetry phenotypes (Scheiner et al., 1991; Clarke, 1995; Zakharov et al., 2020). Fluctuating asymmetry (FA) refers to small, random deviations from bilateral symmetry that indicate developmental instability in bilaterally symmetrical organisms (Vangestel et al., 2011; Klingenberg, 2015). Increases in asymmetry can impact resource acquisition, sexual selection, reproduction, and individual survival (Palmer & Strobeck, 1986; Graham et al., 2010), potentially serving as an early warning signal for ecosystem-level impacts. If elevated temperature represents a source of developmental stress for organisms, we expect individuals to exhibit higher levels of FA. Moreover, the greater the temperature variation at a given location, the greater the developmental stress between individuals inhabiting ‘natural’ and warmer environments. Therefore, a relationship between rising temperature and the level of asymmetry in ectothermic organisms could indicate increased metabolic costs due to stress affecting cellular restoration.

Aquatic invertebrates are ideal study organisms, as their early life stages are relatively immobile compared with larger vertebrates (e.g., fish), reducing their capacity to avoid unfavorable conditions. In contrast, adult stages are often capable of flight,

which confers high mobility and facilitates dispersal among aquatic systems. They also play an important role in aquatic ecosystems, with some acting as predators and exerting significant influence on ecosystem processes, such as top-down control (Rocha et al., 2012). Representative examples are aquatic hemipterans (Vianna & Melo, 2003), which are predominantly predators feeding on a variety of prey such as insects, cladocerans, amphipods, tadpoles, and small fish (Ohba, 2019). Their hemimetabolous development, characterized by incomplete metamorphosis in which nymphs resemble adults, facilitates the assessment of morphological changes, such as asymmetry associated with metabolic alterations during growth.

In this study, we sampled hemipterans from tropical geothermal and non-geothermal streams to investigate how increased temperatures influence community structure, morphology, and bilateral symmetry. Our objective was to understand the relationship between temperature and Hemiptera abundance, richness, proportion of nymphs and adults, body size, and especially bilateral symmetry as metrics to analyze the consequences of temperature increases. We tested the following hypotheses: (I) In environments with higher temperatures, we expect that the richness and abundance of both young and adult aquatic hemipterans will decrease. Environmental temperature will directly impact the metabolism and physiological functions of these ectotherms, potentially increasing individual energy expenditure and reducing energy allocation for reproduction, thereby decreasing the abundance and richness of more sensitive species (Brown et al., 2004; Kozłowski et al., 2004; Bernhardt et al., 2018). (II) Aquatic hemipterans will be smaller in streams with higher temperatures, given the sensitivity of ectothermic insects to these thermal variations. All else being equal (e.g., resources), with an accelerated metabolism, a larger portion of energy is allocated to basal metabolism, restricting resources for somatic growth (Kozłowski et al., 2004; Savage et al., 2004). (III) Aquatic hemipterans developing in streams with higher temperatures are more susceptible to exhibiting fluctuating asymmetry, resulting in significant differences in the bilateral symmetry of their bodies. Energy trade-offs under thermal stress hinder proper morphological development,

increasing developmental instability (Clarke, 1995; Hogg & Williams, 1996; Graham et al., 2010).

Methods

Study area

Field sampling was conducted in October 2022 across 18 streams (Fig. 1), primarily ranging from first- to third-order, located in the southeastern region of the state of Mato Grosso, Brazil. The study area encompasses portions of two hydrographic basins, Tocantins–Araguaia and Paraguay, including the municipalities of Barra do Garças, General Carneiro, Jaciara, Juscimeira, and Santo Antônio do Leverger. The region is influenced by geothermal activity associated with deep groundwater circulation and elevated regional heat flow. In this system, groundwater infiltrates through fracture networks, circulates at depth, and is heated by the geothermal gradient before re-emerging at the surface, contributing to the warming of stream waters (Godoy & Godoy, 2016). As a result, the sampled streams exhibited a pronounced thermal gradient, with water temperatures ranging from 20.60 to 40.26 °C (mean $\pm$ SD = 29.97 $\pm$ 5.27 °C). Aside from temperature, streams showed similar values for physical habitat structure metrics (Fig. 2), indicating that thermal conditions were the main environmental factor varying among sites.

Data collection and sampling design

In each stream, sampling was conducted along a 30-m transect, subdivided into 5-m intervals. At each interval, aquatic invertebrates were collected using a Surber sampler (250 μ m mesh, 30 $\times$ 30 cm sampling area), totaling six subsamples per transect. This was complemented with D-net sampling for 20 s at each interval along the stream margins. Additionally, electrofishing was conducted using a continuous current generator set at 750 V. This procedure was carried out by a three-person team: two operators with dip nets and one responsible for the generator. A single electrofishing pass was performed per 30-m transect. While focused on sampling fish, hemipterans sampled with this method were also considered in further analyses. To characterize stream conditions, we measured water temperature, pH, conductivity, dissolved

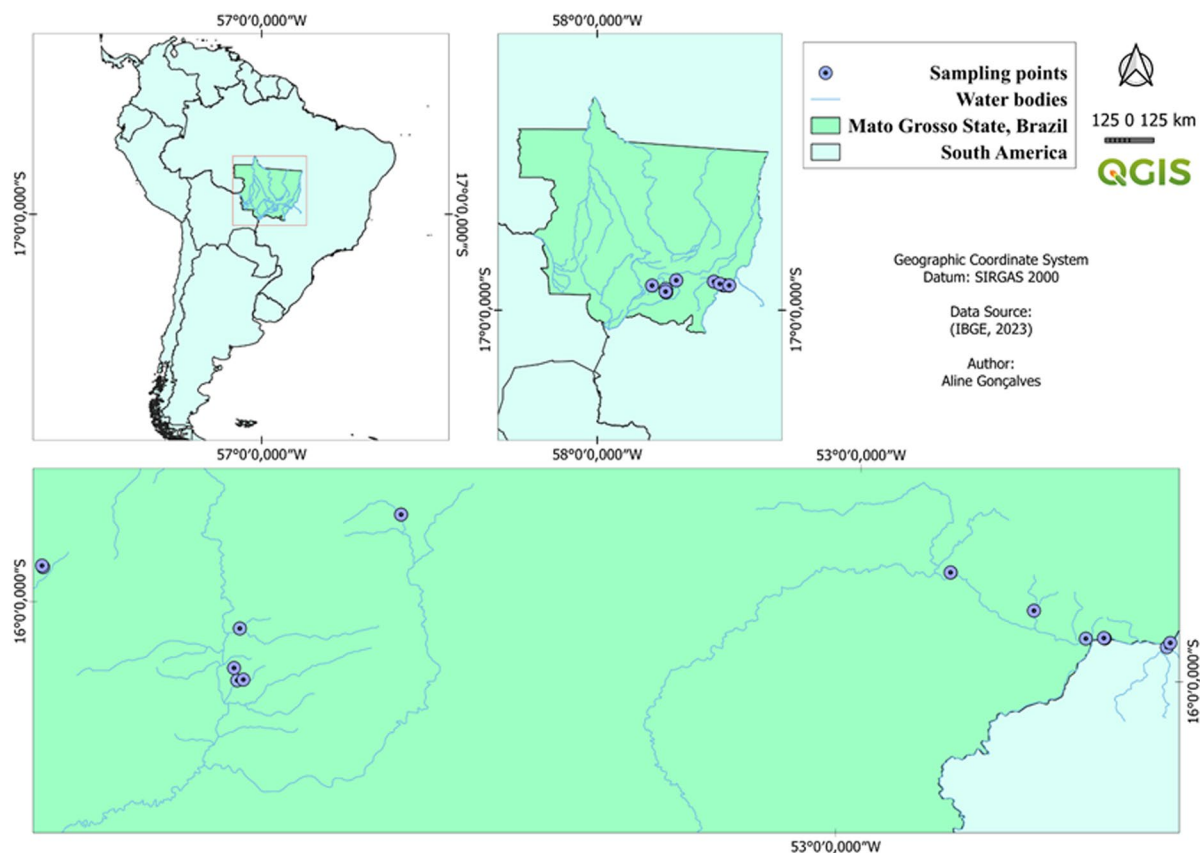


Fig. 1 Map showing the geographic distribution of the 18 stream sampling sites in the state of Mato Grosso, Brazil

oxygen (DO), and total dissolved solids (TDS) using a HORIBA® U-50 multiparameter probe. Measurements were taken at three equidistant points along each transect. Physical habitat variables such as channel width and depth were also recorded every three meters along the 30-m stretch.

Physical and chemical structure of streams

Considering that we focused on understanding the main effect of temperatures on aquatic Hemiptera, we investigated the correlation between the physical and chemical structure of streams along the temperature gradient to understand potential confounding effects. First, we conducted a Principal Component Analysis (PCA) with the following variables: pH, conductivity, dissolved oxygen, temperature and total dissolved solids (Fig. 2). The aim was to explore the environmental variation present in the collected streams and understand if other variables were strongly correlated

with temperature. The first two axes of the Principal Component Analysis (PCA) explained 56.94% of the variation in our data (Fig. 2). The first axis describes the temperature gradient, with warmer streams tending to be wider and deeper. However, given the moderate explanatory power of this axis, we further examined the relationship between stream temperature and physical structure through linear regressions. We found no evidence that width ($F = 3.64$, $P = 0.07$) or depth ($F = 1.81$, $P = 0.19$) was associated with environmental temperature. Therefore, we conclude that stream temperature, our primary variable of interest, was not strongly associated with other physical or chemical variables, minimizing the risk of confounding effects in subsequent analyses.

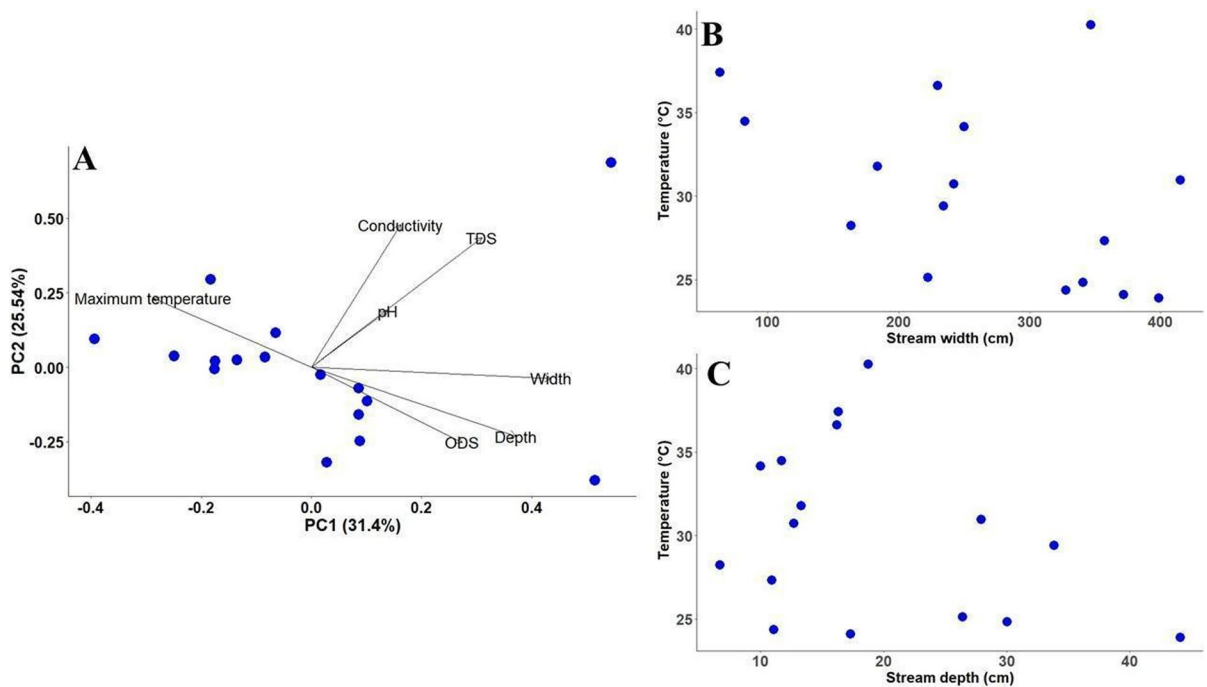


Fig. 2 **A** Principal component analysis (PCA) of abiotic variables across the 18 stream sampling sites in Mato Grosso, Brazil. **B** Relationship between stream width and temperature. **C**

Relationship between stream depth and temperature. Points represent individual sampling sites along a thermal gradient

Specimen identification

Aquatic hemipterans collected through all three sampling methods were sorted alive in the laboratory and subsequently preserved in 70% ethanol. All specimens were identified to the lowest possible taxonomic level under a stereomicroscope, using specialized identification keys (Hamada et al., 2014; Hamada et al., 2018).

Morphological data and measurements

Data analysis of richness, abundance, proportion of nymphs and adults and effects on symmetry

To test part of our first hypothesis, we applied Linear Regression Models (LMs) to relate richness, total abundance, and proportion of nymphs and adults of each stream (response variables) with the average temperature of the streams (explanatory variable). When the same species was found along the temperature gradient (e.g., was found in different sites), we used linear regressions to assess if the abundance

of each specific species decreased with increasing temperature.

To test part of hypotheses II and III, aimed at exploring the effects of temperature on size and symmetry, data were obtained through photographic captures taken under a stereomicroscope with a millimeter ruler. All morphological measurements were performed in triplicate using IC Measure® software (version 3.0.0.503) to minimize measurement error, and the mean value of the three measurements was used in all subsequent analyses. The recorded dimensions for all individuals included Total Length (TL), Total Width (TW), Interocular Distance (ID), and Standard Length (SL). In addition, the total lengths of each leg raptorial, accessory, and swimming leg, as well as the lengths of the tibiae and tarsi on both left and right sides were measured. Wing dimensions were also recorded, including length and width for both sides (Fig. 3).

Fluctuating Asymmetry (FA) was calculated following the approach proposed by Palmer and Strobeck (1994), using the formula: $FA : \sum [(R - L) / (R + L)] / 2$. Here, R (right)

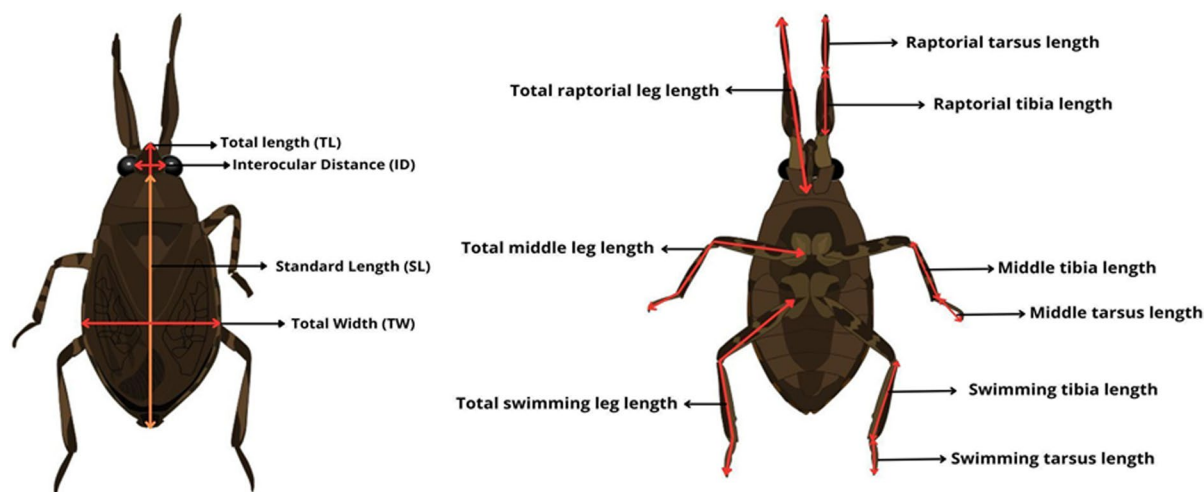


Fig. 3 Schematic drawing of an individual of the species *Belostoma anurum* (Herrich-Schäffer, 1848) demonstrating the name, location, and nomenclature of the measurements taken on the aquatic invertebrates

and L (left) indicate the measurements of a specific dimension paired for the same individual. We conducted measurements on a sample consisting of 246 aquatic Hemiptera, with a specific focus on analyzing size measurements (total and standard), widths, and interocular distance in relation to temperature, using size-corrected morphometric indices to account for overall body size effects. However, due to wing damage, only 115 wing pairs could be measured reliably. Ensuring the validity of the conclusions, we restricted our analysis to adults and controlled for random effects, such as the influence of region and morphotype on models (see below).

To investigate morphometric variation along the thermal gradient, we analyzed body size and shape-related traits using generalized linear mixed models (GLMMs). Temperature was included as a fixed effect, and Region and Morphotype were included as random intercepts to account for spatial structure and morphotype-level variation. Total body length was modeled directly. To control for overall body size, we also analyzed size-corrected morphometric indices, calculated as the ratios of interocular distance/total length, total width/total length, and standard length/total length. These models assumed a Gaussian error distribution. Fluctuating asymmetry responses were analyzed using GLMMs with a Tweedie distribution and log link, as FA values are non-negative. To account for multiple testing in the analyses addressing hypotheses II and III, Bonferroni corrections

were applied separately to each family of related comparisons, with statistical significance determined at adjusted $P < 0.05$. Model adequacy was evaluated through residual diagnostics to ensure appropriate model fit. All analyses were conducted in the R statistical environment, version 4.4.2 (R Core Team, 2024), using the packages dplyr (Wickham et al., 2023), ggplot2 (Wickham, 2016), ggfortify (Tang et al., 2016), glmmTMB (Brooks et al., 2017), and DHARMa (Hartig, 2022).

Results

Abundance, richness, and proportion of adults and nymphs

During our sampling, a total of 285 aquatic Hemiptera were collected, which were classified into 12 distinct morphotypes belonging to the families Belostomatidae and Naucoridae (Table S1). Species richness was negatively associated with temperature ($F = 6.61$, $R^2 = 0.29$, $df = 4.81$, $P = 0.02$), whereas no significant relationship was observed for total abundance ($F = 2.73$, $R^2 = 0.15$, $df = 5.34$, $P = 0.11$). Therefore, we conducted a specific investigation on the abundance of the three most common species. We observed a negative relationship between abundance and temperature for *Limnocoris burmeisteri* De Carlo,

1967 ($F = 14.24$, $R^2 = 0.75$, $df = 3.07$, $P < 0.001$), but not for *Belostoma anurum* (Herrich-Schäffer, 1848) ($F = 0.84$, $R^2 = 0.05$, $df = 5.63$, $P = 0.37$), or *Pelocoris* sp. ($F = 0.16$, $R^2 = 0.01$, $df = 5.74$, $P = 0.68$). Most individuals were recorded at sites with lower temperatures, particularly for *Limnocoris* species, whereas only a few taxa were represented at the warmest sites.

We observed a surprisingly low number of nymphs for the sampled hemipteran groups ($N = 39$), representing only 13.68% of the total individuals. Among the species with the highest number of nymphs, *Belostoma anurum* showed relatively similar numbers across the thermal gradient, with 17 nymphs in the warmest sites and 13 in the coolest ones (Table S1). In contrast, adult individuals were strongly concentrated at the lower-temperature sites, with a total of 246 adults, of which 27 occurred at the warmest sites and 219 at the coolest ones.

Effects on morphology

Morphometric measurements of the collected specimens varied widely; however, there were no significant differences in the adult body sizes of individuals collected from streams of different temperatures (Table 1, Fig. 4). The total length of the invertebrates ranged from 0.521 cm to 4.06 cm, with an average size of 1.06 cm. The smallest individuals were nymphs from environments with mild temperatures ranging from 20 to 25 °C.

Fluctuating asymmetry

Overall, the results indicate that increasing temperature is associated with higher levels of fluctuating asymmetry (FA) in adult aquatic hemipterans,

particularly in locomotor and flight-related structures (Table 2; Figures S1 and S2). When leg-related traits were considered as whole structures, raptorial and middle legs showed no significant relationships with temperature (raptorial legs: $Z = 2.112$, $P = 0.3817$; middle legs: $Z = 2.610$, $P = 0.0997$), whereas swimming legs exhibited a significant positive association between FA and temperature ($Z = 2.838$, $P = 0.0499$). A more detailed analysis of leg segments showed that no significant effects were detected for the tibial segments of raptorial, middle, or swimming legs (raptorial legs: $Z = 1.928$, $P = 0.5921$; middle legs: $Z = 1.996$, $P = 0.5056$; swimming legs: $Z = 2.305$, $P = 0.2326$). Similarly, no significant effects were detected for the tarsi of raptorial and swimming legs (raptorial legs: $Z = 1.863$, $P = 0.6872$; swimming legs: $Z = 1.456$, $P = 1.0000$). In contrast, temperature was positively associated with FA in the tarsi of middle legs ($Z = 3.453$, $P = 0.0061$). Wing length asymmetry also increased significantly with temperature ($Z = 2.850$, $P = 0.0481$; Table 2; Figures S1 and S2), whereas wing width asymmetry could not be reliably evaluated at the community level due to model convergence.

In addition to the general analysis of the twelve morphotypes collected along the geothermal gradient, we conducted specific analyses (linear regressions) using adults of *B. anurum* and *L. burmeisteri* (Table 3; Figures S3 to S5), which were selected for showing more pronounced responses to elevated temperatures. For *B. anurum*, we did not observe significant differences in general body morphometric measurements in relation to temperature, such as total length ($P = 0.566$), interocular distance ($P = 1.000$), total width ($P = 0.580$), and standard length ($P = 1.000$). Likewise, no

Table 1 Results of the generalized linear mixed models (GLMMs) fitted with the package glmmTMB to assess the relationship between the temperature gradient (fixed effect) and morphometric traits of adult aquatic hemipterans. A total of 246 individuals were analyzed, with Morphotype (12 levels) and Region (3 levels) included as random effects. Morphometric traits were modeled assuming a Gaussian error distribution. Reported P values were adjusted for multiple comparisons using the Bonferroni correction

Measures	Estimate	Standard error	Z	AIC	BIC	LogLik	Bonferroni-adjusted P
Total length (TL)	0.010541	0.007753	1.359	203.2	220.8	-96.6	0.696
Interocular distance (ID)	-0.00005	0.0004	-0.304	-1410.0	-1392.4	710.0	1.000
Total width (TW)	0.0003868	0.0009976	0.388	-821.7	-804.2	415.9	1.000
Standard length (SL)	-0.0004063	0.0009274	-0.438	-867.9	-850.4	439.0	1.000

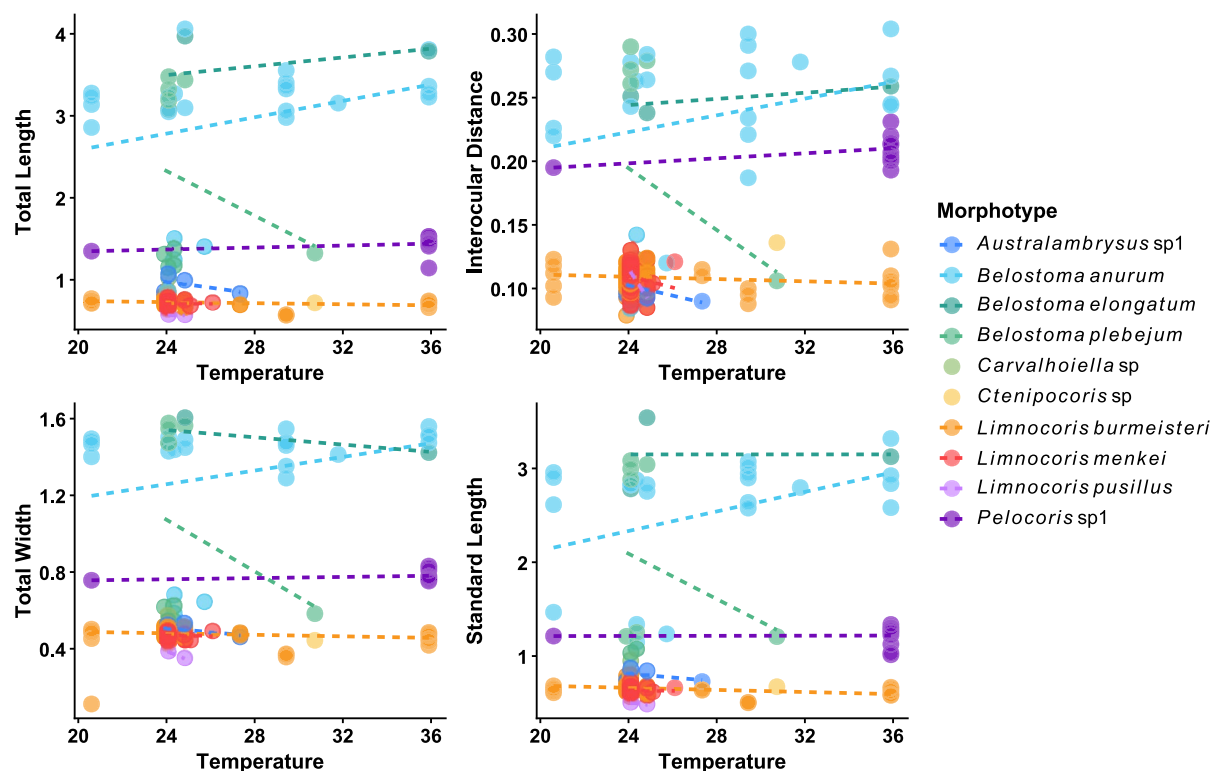


Fig. 4 Relationship between morphological characteristics of adult aquatic hemipterans (total length, interocular distance, total width, and standard length) and stream temperature along a thermal gradient. Relationships were evaluated using Generalized Linear Mixed Models (GLMMs) fitted with the

glmmTMB package, with Region and Morphotype included as random effects. No significant relationships were detected between temperature and any of the morphological traits ($p > 0.05$ for all models). Dashed lines represent the overall fitted relationships

significant relationships between temperature and fluctuating asymmetry were detected for the raptorial legs ($P = 0.358$), raptorial tibiae ($P = 0.143$), raptorial tarsi ($P = 0.564$), or swimming tarsi ($P = 0.370$). Structures related to flight, such as wing length and width, remained significantly associated with temperature ($P = 0.002$ for both) (Figures S3 to S5), suggesting that warmer environments may compromise the symmetrical development of these traits.

In the case of *L. burmeisteri*, the effects of rising temperature were particularly evident. We recorded a significant reduction in total length ($P = 0.003$) and standard body length ($P < 0.001$), while body width and interocular distance did not show relevant changes (Figures S3 to S5). Pronounced alterations were also observed in locomotor appendages and wings, with increased asymmetry in the measurements of raptorial legs ($P < 0.001$), swimming legs

($P < 0.001$), middle leg tibiae ($P < 0.001$), swimming leg tibiae ($P < 0.001$), raptorial tarsi ($P < 0.001$), middle leg tarsi ($P = 0.026$), as well as wing length ($P < 0.001$) and width ($P < 0.001$) (Figures S3 to S5). These findings reinforce that overall body size as well as the symmetrical development of functional structures can be markedly altered in response to thermal stress in some species.

Discussion

In this study, we investigated how increased temperature influences the community structure, morphology, and bilateral symmetry of aquatic hemipterans. We hypothesized that higher temperatures would reduce abundance and richness, decrease body size, and increase fluctuating asymmetry due to thermal stress and energetic trade-offs during development. To

Table 2 Results of the generalized linear mixed models (GLMMs) fitted with the glmmTMB package to assess the relationship between the temperature gradient (fixed effect) and fluctuating asymmetry (FA) of adult aquatic hemipterans. A total of 246 individuals were analyzed for body asymmetry traits, while 115 individuals were included in the analysis of wing asymmetry. Morphotype (12 levels) and Region (3 levels) were included as random effects. FA responses were modeled using a Tweedie distribution with a log link. Reported P values were adjusted for multiple comparisons using the Bonferroni correction

Measures	Estimate	Standard error	Z	Bonferroni-adjusted P
Raptorial legs	0.3330	0.1577	2.112	0.3817
Middle legs	0.4095	0.1569	2.610	0.0996
Swimming legs	0.4729	0.1666	2.838	0.0498
Raptorial tibiae	0.9862	0.5114	1.928	0.5920
Middle leg tibiae	0.4092	0.2050	1.996	0.5056
Swimming tibiae	0.4786	0.2076	2.305	0.2326
Raptorial tarsi	0.2842	0.1525	1.863	0.6872
Middle leg tarsi	0.6819	0.1975	3.453	0.0060
Swimming tarsi	0.3223	0.2213	1.456	1.0000
Wing length	0.8253	0.2896	2.85	0.0480
Wing width	1.226	NaN	NaN	NaN

Bold values indicate statistically significant P-values

test these predictions, we sampled eighteen streams, including geothermal and non-thermal systems, in the Central-West region of Brazil using a natural experimental design. We observed that geothermal streams exhibited lower abundances of aquatic hemipterans and reduced species richness compared to non-thermal environments, supporting our first hypothesis. At the species level, *Limnocoris burmeisteri* exhibited the strongest responses to the thermal gradient. Regarding morphology, we did not detect general differences in mean body size between environments; however, species-specific analyses revealed that *L. burmeisteri* showed a reduction in body size along the gradient, partially supporting our second hypothesis. Finally, we observed a marked increase in overall fluctuating asymmetry, particularly in the wings, suggesting that thermal stress associated with higher temperatures may induce developmental instability, in line with our third hypothesis.

Our first hypothesis was only partially supported, as temperature did not produce the expected effect on the total abundance of aquatic hemipterans.

However, when analyses were conducted at the species level, we observed a pronounced reduction in the number of individuals in geothermal environments for some species. This pattern is consistent with previous studies reporting declines in aquatic invertebrate abundance under warming conditions (Daufresne et al., 2004; Durance & Ormerod, 2007; Heino et al., 2009; Dossena et al., 2012; Quenta-Herrera et al., 2021; Jorgensen et al., 2022). When directly analyzing richness, we detected a decrease in the number of species in geothermal streams, likely reflecting the thermal tolerance limits of aquatic hemipterans (Brown et al., 2004; Zivic et al., 2014; Saito et al., 2021; Jorgensen et al., 2022). The lower representation of families and genera in thermal streams follows the patterns described in other geothermal systems, particularly in volcanic regions (Duggan et al., 2007; Clements et al., 2011), as reinforced by subsequent studies (Zivic et al., 2013; Loyola et al., 2014; Quenta-Herrera et al., 2021). In our study, for example, three species of the genus *Limnocoris* were recorded in non-thermal streams, whereas only one occurred in heated environments.

We observed distinct responses in the distribution of developmental stages along the thermal gradient. Adults were less frequent in warmed streams, whereas nymphs occurred in similar proportions in both thermal and non-thermal environments. This pattern suggests that although hemipterans are able to colonize warmer habitats during early life stages, they face physiological or ecological constraints that hinder the transition to the adult stage. This mismatch may reflect both negative effects of high temperatures on survival or development of immature stages (Pritchard, 1991; Brown et al., 2004) and, for species capable of active flight, dispersal toward cooler areas in search of more favorable conditions (Cozma et al., 2020).

The observed population-level patterns were also reflected in morphological responses. Warmer environments impose metabolic constraints on aquatic invertebrates, potentially leading to changes in development and body size, as predicted by metabolic theory (Brown et al., 2004). Our results only partially meet this expectation, as a significant reduction in body size along the thermal gradient was observed for only one of the evaluated species. Nevertheless, this response is relevant, since warming was also

Table 3 Results of linear models (LMs) evaluating the relationship between temperature and morphometric and symmetry variables in *Belostoma anurum* and *Limnocoris burmeisteri*. Models were fitted using ordinary least squares, with temperature included as a fixed predictor. Response variables correspond to asymmetry in different leg segments (total,

middle, and distal), as well as morphometric measurements (length, width, and distance between body structures). Model diagnostics and summary statistics are provided, including estimates, standard errors, t-values, and Bonferroni-adjusted *p*-values

	Measures	Estimate	Standard error	t-value	Bonferroni-adjusted P
<i>Belostoma anurum</i>	Total length (TL)	0.04989	0.03273	1.524	0.566
	Interocular distance (ID)	−0.00005	0.0004	−0.304	1.000
	Total width (TW)	0.0003868	0.0009976	0.388	0.580
	Standard length (SL)	−0.0004063	0.0009274	−0.438	1.000
	Raptorial legs	0.0024	0.0010	2.281	0.358
	Middle legs	0.0023	0.0012	1.864	0.832
	Swimming legs	0.0019	0.0010	1.835	0.880
	Raptorial tibiae	0.0013	0.0005	2.700	0.143
	Middle leg tibiae	0.0011	0.0008	1.457	1.000
	Swimming tibiae	0.0025	0.0012	2.040	0.588
	Raptorial tarsi	0.0041	0.0020	2.061	0.564
	Middle leg tarsi	0.0014	0.0017	0.796	1.000
	Swimming tarsi	0.0041	0.0018	2.265	0.370
	Wing length	0.0037	0.0008	4.556	0.002
	Wing width	0.0107	0.0023	4.533	0.002
	Measures	Estimate	Standard error	t-value	Bonferroni-adjusted P
<i>Limnocoris burmeisteri</i>	Total length (TL)	−0.0032	0.0012	−2.679	0.003
	Interocular distance (ID)	7.563e−05	3.763e−04	0.199	1.000
	Total width (TW)	0.0001	0.0015	0.084	1.000
	Standard length (SL)	−0.0036	0.0009	−4.041	<0.001
	Raptorial legs	0.0016	0.0002	7.447	<0.001
	Middle legs	0.0002	0.0001	2.738	0.077
	Swimming legs	0.0002	0.0003	6.873	<0.001
	Raptorial tibiae	0.0009	0.0003	2.546	0.132
	Middle leg tibiae	3.258e−04	7.472e−05	4.361	<0.001
	Swimming leg tibiae	1.377e−04	3.337e−05	4.127	<0.001
	Raptorial tarsi	7.30E−01	9.201e−05	7.928	<0.001
	Middle leg tarsi	0.0030	0.0009	0.0009	0.026
	Swimming tarsi	0	0	NaN	NaN
	Wing length	0.0037	0.0004	7.695	<0.001
	Wing width	0.0086	0.0009	9.404	<0.001

Bold values indicate statistically significant P-values

associated with reduced species richness, indicating the action of an environmental filter that eliminates less tolerant species. Among those that persist, particularly the most abundant ones, body size reduction may represent a compensatory strategy in response to increased metabolic demands imposed by thermal

stress (Jourdan et al., 2019; Verberk et al., 2020; Bonacina et al., 2022).

In addition to body size, elevated temperatures may also affect more subtle aspects of invertebrate morphology. In high-temperature environments, aquatic invertebrates may experience reductions in body size (Atkinson, 1994; Forster et al., 2012; Saito

et al., 2021), as well as changes in bilateral symmetry (Graham et al., 2010; Klingenberg, 2015). Organisms are genetically predisposed to develop phenotypes regulated by homeostatic mechanisms that buffer developmental “noise,” ensuring stable morphological expression (Lerner, 1954; West-Eberhard, 2003). However, environmental stressors acting during critical developmental windows may disrupt these regulatory mechanisms, resulting in increased fluctuating asymmetry (Klingenberg, 2001, 2015). In our study, fluctuating asymmetry was detected in the legs and wings of aquatic hemipterans, with a significant increase along the thermal gradient. These structures play essential roles in locomotion, dispersal, foraging, and reproduction (Polhemus, 2007; Ohba, 2019), and asymmetries in them may have ecological and fitness consequences, reducing flight efficiency, increasing energetic costs, and compromising predator avoidance and reproductive success (Møller, 1996; Beasley et al., 2013; Corrêa-Silva et al., 2024).

Previous studies have linked increased fluctuating asymmetry to different environmental stressors in aquatic ecosystems, such as flow variation, pesticide contamination, eutrophication, and nutritional stress (Clarke, 1990; Tsubaki, 1998; Labrie et al., 2003; McLachlan, 2010; Beasley et al., 2013; Corrêa-Silva et al., 2024). Thermal stress has also been associated with developmental instability in stream invertebrates, with experimental evidence showing increased fluctuating asymmetry under warming conditions (Hogg & Williams, 1996). However, Hogg et al. (2001) did not detect increased fluctuating asymmetry under experimental warming when using meristic traits (leg spine counts). In contrast, the present study detected significant responses using continuous morphometric traits, suggesting that the type of morphological trait evaluated may influence the detection of developmental instability under thermal stress. Although many freshwater ecosystems are already experiencing the effects of thermal variability, such as biodiversity loss and trophic restructuring (Woodward et al., 2010), these impacts may be underestimated when intra-individual morphological metrics are not incorporated. Although environmental stress can generate selective pressures over time (Hoffmann & Sgrò, 2011), high levels of fluctuating asymmetry

generally indicate developmental instability rather than improvements in performance or fitness.

A more detailed, species-level assessment revealed contrasting responses to high temperatures in *Belostoma anurum* (Belostomatidae) and *Limnocoris burmeisteri* (Naucoridae), reinforcing that thermal stress can affect organisms differently, even among functionally similar aquatic predators (Merritt et al., 2008). *Limnocoris burmeisteri* showed a pronounced reduction in body length with increasing temperature, accompanied by an increase in fluctuating asymmetry. These changes indicate that thermal stress may compromise normal ontogenetic development, directly affecting functional performance such as predatory efficiency, locomotion, and predator avoidance (Parsons, 1990; Piscart et al., 2005). In contrast, *Belostoma anurum* did not show significant changes in body size. A possible explanation is related to the reproductive biology of belostomatids, in which adult males carry eggs attached to their dorsal surface throughout embryonic development (Smith, 1997). Because the available dorsal space represents an important and potentially limiting reproductive resource (Kraus, 1989), selection may favor the maintenance of sufficient dorsal surface area, potentially constraining reductions in body size even under thermal stress. Nevertheless, *B. anurum* exhibited pronounced asymmetry in wing traits, suggesting that developmental stability may be affected even when overall body size remains unchanged. Wing asymmetry may negatively affect dispersal and efficient habitat use, especially in the adult stage, when wing integrity is essential for movement among water bodies (Pinto et al., 2012; Ohba, 2019).

Studies on tropical macroinvertebrates indicate that loss of symmetry is a sensitive indicator of environmental stress, particularly thermal stress (Savage & Hogarth, 1999; Hoffmann et al., 2003; Chang et al., 2007), and in tropical environments, asymmetry may signal the approach or exceedance of physiological limits (Bonebrake & Deutsch, 2012; Colinet et al., 2015; Kingsolver et al., 2015). Additional evidence shows that warming tends to reduce body size in aquatic ectotherms, particularly in predatory species, due to changes in energy allocation under suboptimal conditions (Dell et al., 2011; Forster et al., 2012; Sentis et al., 2024). In the case of *L. burmeisteri*, this

pattern is confirmed, reinforcing the vulnerability of aquatic predators to thermal changes and their potential as sensitive bioindicators of environmental alteration (Woodward et al., 2010).

Conclusion

This study examined how abundance, richness, body morphology, and symmetry of aquatic hemipterans respond along a natural geothermal gradient. Higher temperatures were associated with changes in abundance and richness, shifts in the proportion of juveniles and adults, and increased fluctuating asymmetry in some morphological traits, indicating widespread effects of temperature changes in aquatic hemipterans. No general reduction in body size was detected among the studied hemipterans; instead, a decrease in body size along the thermal gradient was restricted to a single species. Additionally, individuals from warmer environments exhibited pronounced fluctuating asymmetry, which may impair feeding, reproduction, and overall performance, potentially affecting their persistence in aquatic systems. In the context of increasing climate change, our findings provide novel insights into the sublethal effects of heat stress on aquatic predators and highlight the importance of considering morphological instability when assessing the impacts of warming on Neotropical freshwater ecosystems.

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Data availability Data used in this work were deposited in a permanent repository in accordance with the instructions for authors: <https://doi.org/10.5281/zenodo.19292012>.

Declarations

Conflicts of interest The authors declare that they have no conflict of interest related to the publication of this manuscript.

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